



Herbal nanoparticles: bridging traditional medicine and modern science in epilepsy treatment

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Abstract

Millions of people worldwide suffer from epilepsy, a persistent neurological illness characterized by recurring seizures. Despite advances in anti-epileptic medications, a significant number of patients continue to have insufficient seizure control and have side effects. Nanotechnology has emerged as a promising alternative for increasing medicine delivery and therapeutic effects in recent years. The anti-epileptic potential of nanoparticles produced from herbal medicines such as berberine-loaded zein/hyaluronic acid composite, *Cannabis sativa* extract-loaded nanoliposomes and nanostructured lipids, nanostructured lipid vehicle-carried safranal, and cryptolepine solid-lipid NPs is reviewed in this work, which makes use of the synergistic effects of nanotechnology and natural substances. The manuscript presents an overview of the mechanisms underlying the anti-epileptic effects of these nanoparticles, ranging from regulating neurotransmitter systems and ion channels to lowering oxidative stress and inflammation. Preclinical studies using animal models of epilepsy have shown that herbal medicine nanoparticles can reduce seizure activity, prolong seizure latency, and improve cognitive function. The findings presented in this manuscript highlight the remarkable potential of herbal medicine nanoparticles as a novel approach to the management of epilepsy. Continued research and development in this field have the potential to revolutionize epilepsy therapy and improve the quality of life for people suffering from this debilitating condition.

Keywords Anticonvulsants · Epilepsy · Seizures · Neurotransmitter agents · Oxidative stress · Inflammation

Abbreviations

4HNE	4-Hydroxynonenal	CB-Rs	Cannabinoid receptors
Ach	Acetylcholine	CCD	Central composite design
AchE	Acetylcholinesterase	CNS	Central nervous system
Apaf-1	Apoptotic protease activating factor 1	COX-2	Cyclooxygenase-2
ASMs	Antiseizure medications	Cur-SLNs	Curcumin encapsulated in solid-lipid nanoparticles
Bax	Bcl-2-Associated X protein	CUQDs	Copper oxide quantum dots
BBB	Blood–brain barrier	DNA	Deoxyribonucleic acid
Bcl-2	B-cell lymphoma-2	EA	Ellagic acid
BAD	Bcl-2 associated agonists of cell death	EEG	Electroencephalogram
BSA	Bovine serum albumin	EMG	Electromyography
		PO	Erythropoietin
		GABA	γ-Aminobutyric acid receptor
		GABAA	γ-Aminobutyric acid receptor type A
		GFAP	Glial fibrillary acidic protein
		GPx	Glutathione peroxidase
		GR	Glutathione reductase
		GSH	Glutathione
		HA	Hyaluronic acid
		HO-1	Heme oxygenase-1
		Hsp70	Heat shock protein family A
		Hsp1b	Heat shock protein family A member 1

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IL-10	Interleukin 10
IFN- α	Interferon- α
JAK2	Phosphorylated Janus kinase
JNK	C-Jun N-terminal kinase
L-NAME	L-NG-Nitro arginine methyl ester
MAPK	Mitogen-activated protein kinase
MESNPs	<i>Mimusops elengi</i> Silver nanoparticles
MDA	Malondialdehyde
mRNA	Messenger RNA
NADPH	Nicotinamide adenine dinucleotide phosphate
NF- κ B	Nuclear factor- κ B
NLV	Nanostructured lipid vehicle
NREM	Non-rapid eye movement
NO	Nitric oxid
NOS	Nitric oxide synthase
Nrf2	Nuclear factor erythroid-2-related factor 2
NP	Nanoparticle
p53	Tumor protein p53
PDI	Polydispersity index
PC	Pilocarpine
PEG	Poly ethylene glycol
PIP-SPL	Piperine-loaded Spanlastics
PLGA	Poly lactic-co-glycolic acid
PLGA-b-PEG	Poly (lactic-coglycolic acid)-block-poly (ethylene glycol)
PTZ	Pentylene-tetrazole
ROS	Reactive oxygen species
SOD	Superoxide dismutase
SLN-CRYP	Cryptolepine called solid-lipid nanoparticle-cryptolepine
TNA	Tannic acid
TNF α	Tumor necrosis factor alpha

Introduction

Epilepsy

Epilepsy is a chronic neurological disorder characterized by recurring seizures (Rahbardar and Hosseinzadeh 2020). Epilepsy affects around 65 million people globally (Devinsky et al. 2018) and could be caused by a range of factors, including acute brain injuries such as autoimmune conditions, central nervous system (CNS) infections, gene mutations, metabolic disorders, neurotrauma or status epilepticus, and stroke. However, in many cases, the origin of epilepsy remains unidentified (Scheffer et al. 2017). Epilepsy is often accompanied by neurological comorbidities, including anxiety, cognitive deficits, and depression, as well as psychiatric disturbances such as autism spectrum disorders, which can significantly impact the patient's quality of life. The incidence and severity of these comorbidities vary depending

on the underlying cause of epilepsy and the age of onset. Despite extensive research, the pathophysiological mechanisms that lead to the onset and recurrence of epileptic seizures, the development of histopathological correlates, and the comorbidities associated with epilepsy remain largely unresolved (Vezzani et al. 2019).

Epileptic seizure classification

Seizures are categorized based on their point of origin in the brain. They may originate in a specific region (focal seizures), simultaneously involve both hemispheres (generalized seizures), or have an undetermined starting location (unknown onset seizures). Awareness, a measure of consciousness, can be used to classify focal seizures further and differentiate between those with preserved and impaired awareness. The three primary categories of seizures are as follows: non-motor seizures, which include autonomic, behavior arrest, cognitive, emotional, and sensory subtypes; motor seizures, which include automatisms, atonic, clonic, epileptic spasms, hyperkinetic, myoclonic, and tonic; and focal to bilateral tonic-clonic seizures, in which the seizure spreads to both hemispheres. The two types of generalized seizures are non-motor/absence (typical, atypical, myoclonic, eyelid myoclonia) and motor (tonic-clonic, clonic, tonic, myoclonic, myoclonic-tonic-clonic, myoclonic-atonic, atonic, epileptic spasms) (Fig. 1) (Fisher 2017).

Mechanisms of epilepsy

Epilepsy's underlying mechanisms are intricate and multifaceted. Among the mechanisms that cause epilepsy are oxidative stress, inflammation, apoptosis, ion channel disturbance, and an imbalance between inhibitory and excitatory neurotransmission in the brain (Fig. 2). One important mechanism is oxidative stress, which occurs when there is an imbalance between the production of reactive oxygen species (ROS) and the ability of the body to detoxify them. This can lead to damage to cellular components and contribute to the development of seizures (Łukawski and Czuczwar 2023). Inflammation is also a key factor in epilepsy, as it can lead to the activation of immune cells and the release of pro-inflammatory cytokines, which can exacerbate seizure activity (Lach et al. 2022). Oxidative stress and inflammation are closely related mechanisms that frequently intensify each other (Mohammadi Zonouz et al. 2023). Glial fibrillary acidic protein (GFAP) is a cytoskeletal protein found in astroglial cells and brain progenitor cells. Glial cell participation in the development of epilepsy is linked to the release of inflammatory proteins such as interleukins and chemokines, which can increase hyperexcitability. Any change in astrocytes can make newly formed neurons vulnerable to aberrant connections, increasing the likelihood

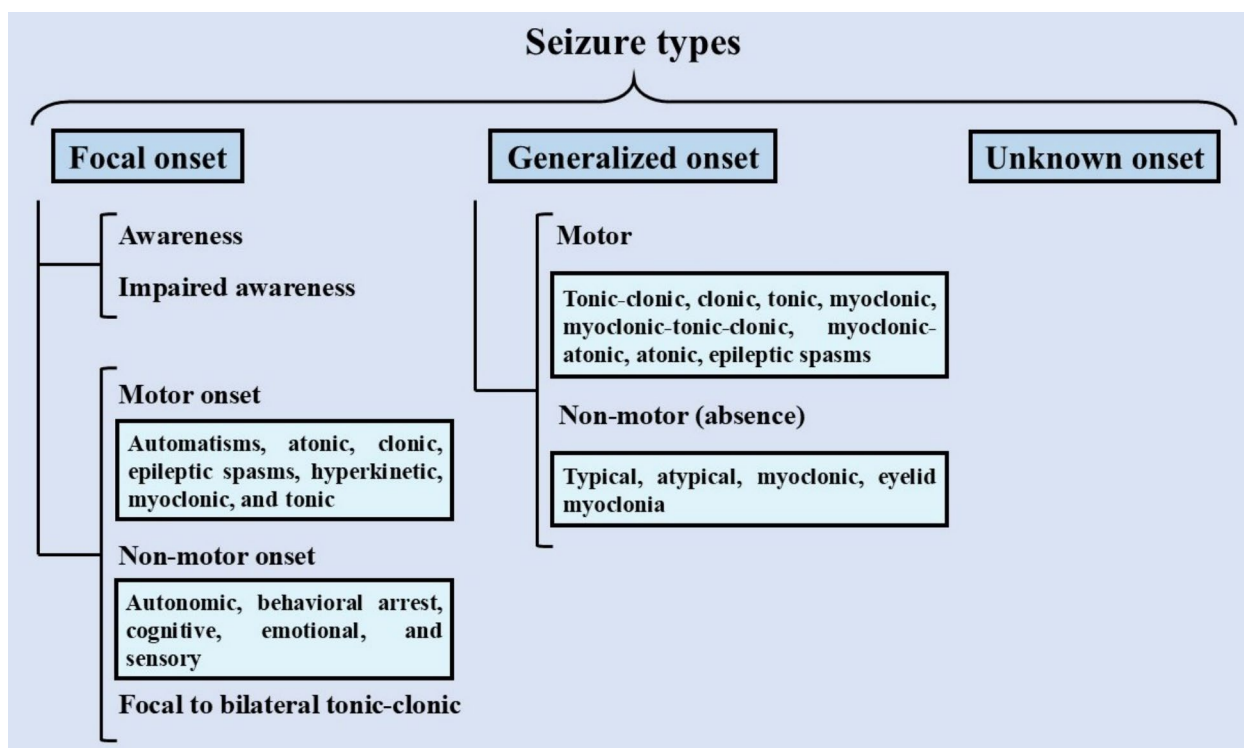


Fig. 1 Types of seizures

of hyper-excitability (Rana and Musto 2018; Sanz and Garcia-Gimeno 2020). Apoptosis, programmed cell death, is another mechanism that has been implicated in epilepsy, as it can lead to the loss of neurons and contribute to the development of seizures (Kakhki et al. 2024; Löscher et al. 2013). Furthermore, ionic channels, which are responsible for regulating the flow of ions across neuronal membranes, have also been implicated in epilepsy. Alterations in the expression or function of these channels can lead to changes in neuronal excitability and contribute to the occurrence of seizures (Bhatia and Bhatia 2016). Moreover, an imbalance between inhibitory and excitatory neurotransmission in the brain, which may be caused by an increase in glutamatergic and/or a decrease in γ -aminobutyric acid (GABA)ergic transmission, is associated with the onset of epilepsy in both human patients and animal models (Sarlo and Holton 2021; van van Hugte et al. 2023).

Antiseizure medications

Overall, understanding the underlying mechanisms of epilepsy is crucial for the development of effective treatments for this disorder. Antiseizure medications (ASMs) are medications used to treat epileptic seizures. While anti-epileptic medications can effectively control seizures in many patients, they can also have significant side effects,

such as dizziness, fatigue, and cognitive impairment. Some ASMs can also cause more serious side effects, such as liver damage or a drop in white blood cell count. Furthermore, ASMs can interact with other medications, making it important for people taking them to be aware of potential drug interactions (Bosak et al. 2019; McNary et al. 2025; Tsvere et al. 2020). Additionally, some patients may not respond to currently available medications, underscoring the need for the development of new ASMs with improved efficacy and tolerability profiles. Improved treatments could help patients achieve better seizure control and minimize the impact of side effects on their quality of life (Adiga et al. 2024; Franco et al. 2016; Lee et al. 2024; Löscher et al. 2013).

Nanoparticles

A nanoparticle (NP) is a minuscule particle with a diameter of less than 100 nm. Despite their small size, NPs function as distinct units in terms of their properties and movement. They are unique because they connect materials at the atomic or molecular level with bulk materials. Given their extensive applications across various industries, research on NPs is currently one of the most actively studied fields (Mody et al. 2010).

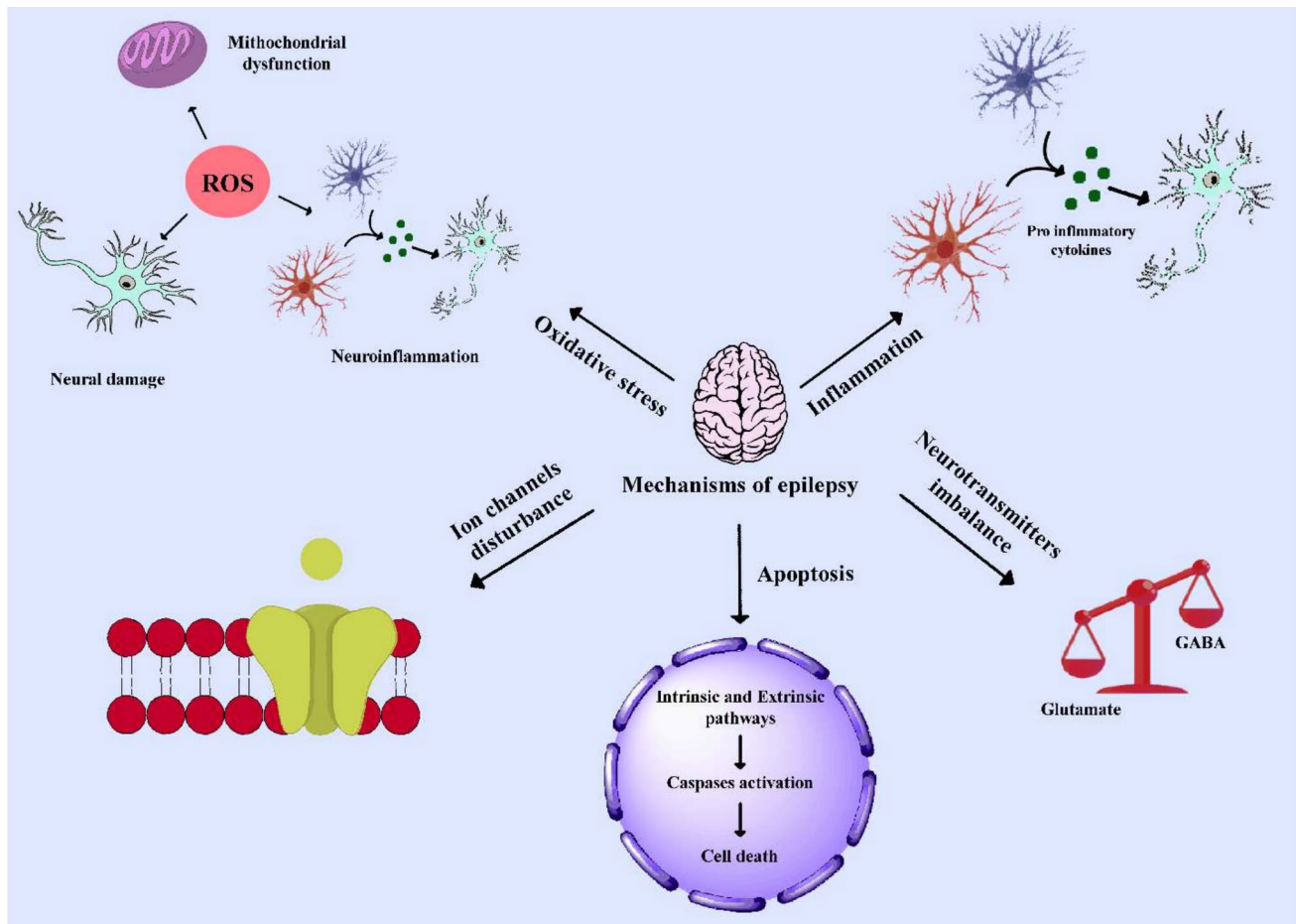


Fig. 2 The underlying mechanisms of epilepsy (oxidative stress, inflammation, neurotransmitter imbalance, apoptosis, and ion channel disturbance)

Classification of NPs

There are different types of NPs such as nanocarriers, nanocrystals, polymeric nanoparticles, nanoemulsions, nanospheres and nanocapsules, metallic NPs, and quantum dots which we will explain each one below. NPs are ultrafine materials important in various applications, most notably medication administration and imaging. They can be classified into different types based on their composition and structure (Bhatia and Bhatia 2016; Guterres et al. 2007; Jalili et al. 2023). Nanocarriers, including liposomes and solid lipid NPs, encapsulate the herbal extract, facilitating its transport across cellular barriers, including the blood–brain barrier (BBB). This improves drug distribution in the brain (Rangsimawong et al. 2016; Sabapati et al. 2019). Nanocrystals increase the solubility and bioavailability of poorly soluble medicines (Chvřla et al. 2024; Gigliobianco et al. 2018). These are pure drug crystals reduced to nano size, enhancing solubility and dissolution rates. This leads to faster and more efficient absorption (Zhou et al. 2017). Polymeric NPs are

adaptable carriers derived from biodegradable polymers. Biodegradable polymers are used to form NPs that can encapsulate the drug, allowing for a sustained and controlled release over time (Khalid and El-Sawy 2017; Kumari et al. 2010) whereas nanoemulsions are stable oil–water combinations that improve medication solubility and increase the bioavailability of lipophilic herbal compounds by enhancing their solubility and stability (Barradas and de Holanda e Silva 2021). Dendrimers are highly branched, tree-like polymers that can carry multiple drugs (Samad et al. 2009). Nanospheres and nanocapsules are polymeric NPs with differing release patterns (Guterres et al. 2007). Metallic NPs, such as gold and silver, have distinct optical and electrical properties, making them valuable in diagnostics (Mody et al. 2010; Mohammadi Zonouz et al. 2024). Lastly, quantum dots are semiconductor NPs that can be utilized for imaging and targeted therapy in drug-delivery systems (Bajorowicz et al. 2018; Gill et al. 2008; Won et al. 2025). Together, these various forms of NPs are revolutionizing the fields of medicine and biotechnology.

Reasons for using NPs in herbal anticonvulsants

By increasing solubility and enabling targeted transport to specific areas in the body, NPs enhance the bioavailability and therapeutic efficacy of herbal anticonvulsants. Their large surface area allows for more effective interactions with biological systems, potentially reducing adverse effects and increasing effectiveness (El-Nahas et al. 2023). NPs offer several advantages in developing herbal anticonvulsants, significantly increasing their therapeutic potential. In the following, we will examine their unique features (Table 1).

First, improved bioavailability: They enhance bioavailability by increasing the solubility and absorption of herbal compounds, which are often hydrophobic and poorly absorbed. This results in more effective therapeutic concentrations in the bloodstream and brain (Matias et al. 2023; Noor et al. 2012).

Second, targeted drug delivery: NPs can be engineered for targeted drug delivery, directing anticonvulsant drugs specifically to areas of the brain affected by seizures. This enhances efficacy while minimizing systemic side effects (Matias et al. 2023).

Third, sustained release: NPs enable sustained release of active ingredients, maintaining therapeutic levels over extended periods and reducing the need for frequent dosing (Hosseini et al. 2011a, b).

Furthermore, reduced toxicity: Encapsulating herbal compounds in NPs can lower potential toxicity, as smaller doses may suffice for desired effects (Matias et al. 2023; Sadeghnia et al. 2020).

A crucial benefit is their enhanced penetration of the BBB: They facilitate transport across the BBB, a significant challenge in treating neurological disorders like epilepsy (Cao and Fang 2006; Movahedpour et al. 2023).

Additionally, overcoming drug resistance: NP formulations can help bypass drug resistance mechanisms that often limit traditional therapies (Sørensen and Kokaia 2013).

Finally, stability and protection: NPs provide stability and protection for sensitive real compounds, ensuring their effectiveness until they reach their target site (Barradas and de Holanda e Silva 2021).

Mechanisms of action of herbal anticonvulsants prepared with NPs

Herbal anticonvulsants, particularly when enhanced with NPs, provide a versatile approach to managing seizures through multiple mechanisms. The first important mechanism is ion channel modulation, where these herbal compounds can affect sodium, potassium, and calcium channels, which are essential for controlling neuronal excitability and preventing seizures (He et al. 2021; Kumar et al. 2019). Second, they contribute to neurotransmitter balance by boosting inhibitory neurotransmitters like GABA and suppressing excitatory ones such as glutamate, thereby stabilizing neuronal activity (El-Missiry et al. 2020; Rajangam et al. 2022). Third, oxidative stress occurs when there is an imbalance between the production of ROS and the body's capacity to detoxify them, leading to cellular damage (He et al. 2021). This condition can result in lipid peroxidation, protein oxidation, and DNA harm, all of which are linked to various health issues, including epilepsy (Scapagnini et al. 2006). Additionally, many herbal extracts possess antioxidant properties that help shield neurons from oxidative stress, a known factor in seizures. Research indicates that NPs made from poly (lactic-co-glycolic acid) containing chrysin can mitigate oxidative stress and neuronal apoptosis while elevating levels of protective proteins like heme oxygenase-1 (HO-1) and nuclear factor erythroid 2-related factor 2 (Nrf2) (He et al.

Table 1 Reason for using NPs in herbal anticonvulsants

Reason for using NPs in herbal anticonvulsants	Description
Improved bioavailability	NPs enhance the solubility and absorption of herbal compounds, leading to more effective therapeutic concentrations in the bloodstream and brain
Targeted drug delivery	NPs can be engineered for targeted delivery, directing drugs specifically to areas of the brain affected by seizures, which enhances efficacy while minimizing systemic side effects
Sustained release	NPs facilitate the sustained release of active ingredients, maintaining therapeutic levels over extended periods and reducing the need for frequent dosing
Reduced toxicity	Encapsulating herbal compounds in NPs can lower potential toxicity, as smaller doses may suffice for desired effects
Enhanced penetration of the blood–brain barrier (BBB)	NPs aid in crossing the BBB, a significant challenge in treating neurological disorders like epilepsy, thus improving drug delivery to the brain
Overcoming drug resistance	NP formulations can help bypass drug resistance mechanisms that often limit traditional therapies, enhancing treatment options
Stability and protection	NPs provide stability and protection for sensitive herbal compounds, ensuring their effectiveness until they reach their target site

2021; Scapagnini et al. 2006). Apoptosis, or programmed cell death (Kakhki et al. 2024), is vital for maintaining neuronal health by removing damaged or non-functional neurons, thus preventing their buildup and supporting overall brain function. During seizures, excessive neuronal activity can activate apoptotic pathways, leading to neuronal loss (Al Omairi et al. 2022; El-Missiry et al. 2020; He et al. 2021). Certain herbal compounds may help manage this process, promoting the survival of healthy neurons while encouraging the death of impaired ones (Challal et al. 2023). Moreover, specific herbal compounds display neuroprotective properties, contributing to neuronal health and safeguarding against damage associated with seizure disorders. The application of NP formulations can enhance the affinity of these herbal compounds for specific neurotransmitter receptors, thereby improving their anticonvulsant effects (Sekhar et al. 2024) (Table 2).

The objective of the research

Throughout history, various types of herbs have been utilized in traditional medicine to treat epilepsy and convulsions. The sixteenth and seventeenth centuries saw the description of 221 plants from 53 plant families, with antiepileptic effects (Adams et al. 2012). Numerous plants have shown anti-epileptic effects in different investigations, including *Rosa damascene* (Hosseini et al. 2011a, b), *Curcuma longa* (Kiasalari et al. 2013), *Nigella sativa* (Noor et al. 2012), *Citrus aurantium* (Hosseini et al. 2011a, b), and *Crocus sativus* (Sadeghnia et al. 2020). The medicinal properties of these herbs can be attributed to their ability to interact with the CNS, receptors such as the γ -aminobutyric acid receptor type A (GABA_A), their anxiolytic or sedative properties (Cao and Fang 2006), and the involvement of neurotrophic factors in restoring the physiological excitability of neurons membrane after a seizure (Barradas and de Holanda e Silva 2021; Sørensen and Kokaia 2013).

The low bioavailability of herbal medicines is a significant challenge in achieving their desired therapeutic effects, as it limits the amount of active compounds that can reach their target site. This is due to a range of factors, including poor solubility, low permeability, rapid metabolism, and limited absorption in the gastrointestinal tract (He et al. 2010). As a result, alternative strategies such as the use of NPs are being explored to improve the bioavailability of herbal medicines (Abdifetah and Na-Bangchang 2019). Synthesizing herbal NPs has the potential to enhance the bioavailability of herbal medicines and enable them to cross the BBB, leading to improved therapeutic outcomes (Zhang et al. 2021a). By reducing the particle size of herbal compounds, their absorption rate can be increased, and their distribution throughout the body can be improved (Chakraborty et al. 2016). Additionally, the use of NPs can allow for targeted drug delivery to specific areas of the body, reducing the potential for side effects and increasing the efficacy of treatment.

The objective of this review is to draw attention to the effective plant NPs that have been studied for their potential use in epilepsy treatment and the potential mechanisms of action underlying the efficacy of these plant NPs in managing epilepsy. Our team also aims to provide a valuable resource for researchers, clinicians, and healthcare professionals in the field. Finally, the goal is to inspire future research, collaboration, and the development of novel plant NP-based medicines that can address unmet clinical requirements in epilepsy care.

Methods

Our team conducted a thorough evaluation of many sources, including Google Scholar, PubMed, Web of Science, and Scopus, to ensure a comprehensive and inclusive analysis. The search included relevant literature in the English language, with no time constraints, and included a wide spectrum of in vitro and in vivo research. The keywords used

Table 2 Mechanism of action of NPs

Mechanism of action	Description
Ion channel modulation	Influences sodium, potassium, and calcium channels to regulate neuronal excitability and prevent seizures
Neurotransmitter balance	Enhances inhibitory neurotransmitters like GABA while suppressing excitatory ones such as glutamate, stabilizing neuronal activity
Oxidative stress reduction	Provides antioxidant protection to neurons, mitigating oxidative stress that can cause cellular damage associated with seizures
Apoptosis regulation	Manages programmed cell death (apoptosis) to promote the survival of healthy neurons and remove damaged ones during seizure activity
Neuroprotective properties	Exhibits neuroprotective effects, contributing to neuronal health and protecting against seizure-related damage
Enhanced delivery across the blood–brain barrier (BBB)	Facilitates the transport of herbal compounds across the BBB, essential for effective treatment of neurological disorders like epilepsy

for this study included "epilepsy," "antiepileptic," "anticonvulsants," "seizures," "herbal medicines," "plant," "natural compounds," "nanoparticles," and "nanoformulation." Figure 3 displays the search process flow chart. The antiepileptic effects of herbal nanoformulations have been introduced as "the name of the plant" and "the name of the other herbal compounds."

Plants

Bacopa monnieri

The plant *Bacopa monnieri*, commonly known as Brahmi a member of the Plantaginaceae family, is a perennial herb traditionally used in Ayurvedic medicine for its cognitive-enhancing properties (Sukumaran et al. 2019). This plant is renowned for its ability to improve memory, reduce anxiety, and enhance overall brain function. The active compounds in *B. monnieri*, particularly bacosides, are believed to support neuronal health by promoting neuroplasticity and protecting against oxidative stress (Rai et al. 2017). Additionally, its anti-inflammatory and antioxidant effects make it a popular choice for supporting mental clarity and focus, contributing to its status as a revered adaptogen in herbal medicine (Valotto Neto et al. 2024).

In the research on poly (lactic-coglycolic acid)-block-poly (ethylene glycol) (PLGA-b-PEG) encapsulated NPs made from *B. monnieri*, scientists examined the anti-seizure

properties of these NPs by conducting both in vitro and in vivo experiments. A kainic acid-induced rat model was used to observe changes in brain activity via electroencephalogram (EEG) and electromyography (EMG). Results indicated that nano-encapsulated bacoside A NPs and nano-encapsulated bacoside I NPs not only maintained over 50% cell viability in neuroblastoma cells but also reduced cytotoxicity when exposed to kainic acid. They effectively decreased the expression of several epileptic markers and protected neurons from apoptosis, while restoring mitochondrial membrane potential. Most importantly, in the rat model, administration of these NPs led to a significant reduction in epileptic spikes, decreased sleep latency, and increased duration of non-rapid eye movement (NREM) sleep. Briefly, bacoside A and bacoside I NPs hold promise in mitigating excitotoxic seizures, potentially providing a novel therapeutic avenue for managing epilepsy and related neurological conditions (Sekhar et al. 2024).

Berberis vulgaris

Berberis vulgaris is a medicinal plant whose primary active compound is berberine, an isoquinoline alkaloid. Berberine exhibits various pharmacological properties, including antioxidant, antimicrobial, antihypertensive, antiarrhythmic, anti-inflammatory, anticancer, antidiabetic, anti-dyslipidemic, anti-obesity, analgesic, and neuroprotective effects (Imenshahidi and Hosseinzadeh 2016; Khoshandam et al. 2022). Berberine has demonstrated the ability to reduce

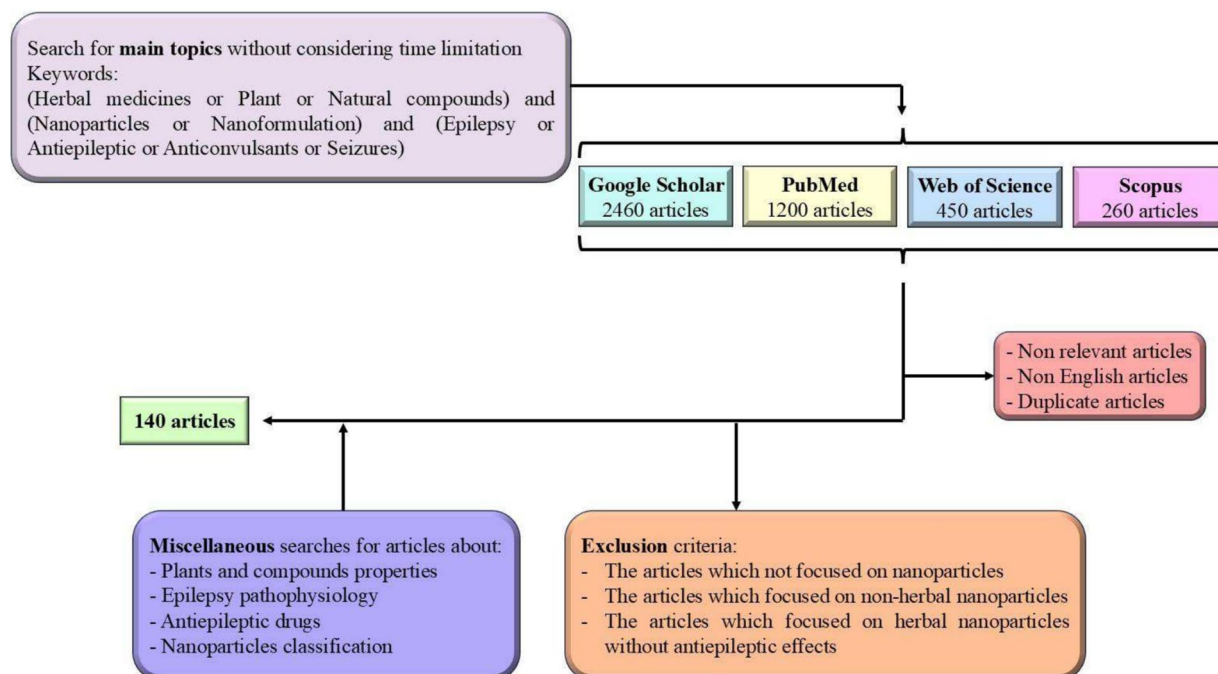


Fig. 3 Flow chart showing the inclusion and exclusion criteria and the search procedure

epilepsy with both short-term and long-term dosing (Zhu et al. 2024). However, because of its extremely low solubility and bioavailability (less than 1%), it has limited clinical usefulness (Khoshandam et al. 2022).

The potential anticonvulsive effect of composite NPs consisting of zein and hyaluronic acid loaded with berberine was investigated in a study. The objective was to facilitate efficient brain uptake of berberine to alleviate neurodegeneration in the pilocarpine (PC) model of epilepsy. These NPs decreased seizure severity, reduced the percentage of animals that developed status epilepticus, and increased the latency of status epilepticus in PC-induced seizure models. The findings indicate that the administration of berberine NPs resulted in a significant increase in the antioxidant activity of the brain as well as a decrease in neurodegeneration, as evidenced by decreased expression of cyclooxygenase-2 (COX-2) and tumor necrosis factor-alpha (TNF- α) (El-Nahas et al. 2023).

Camellia sinensis

Green tea, derived from the unfermented leaves of *Camellia sinensis*, retains a high concentration of its natural compounds. Numerous studies have demonstrated that green tea contains chemical constituents, such as polyphenols, caffeine, and theanine, which contribute to its health benefits. These components can be extracted and have pharmacological effects, including reducing blood sugar levels, providing neuroprotection, and exhibiting anti-cancer properties (Rameshrad et al. 2017; Zhao et al. 2022).

In a study, catechin was loaded into chitosan-coated PLGA NPs, and its bioavailability and anticonvulsant effect were assessed through intranasal administration in the PTZ and increasing current electroshock epilepsy models. The formulation demonstrated a sustained release over 24 h, with improved nasal absorption and prolonged retention for effective distribution from the nose to the brain, all while avoiding first-pass liver metabolism. Furthermore, by using intranasal delivery, chitosan-coated catechin-loaded PLGA NPs were able to effectively alleviate epilepsy at lower doses by crossing the BBB. These NPs also increased the hind limb tonic extension threshold and reduced post-hind limb tonic extension recovery time in increasing current electroshock epilepsy models (Ahmad et al. 2020a, b).

Cannabis sativa

One of the most popular plants for both medical and recreational use is *Cannabis sativa* L. The cannabinoids, terpenoids, nitrogenous chemicals, flavonoids, non-cannabinoid phenols, and steroids are the most significant classes. Pain relief and the treatment of nervous system problems are the main uses of cannabis in medicine. Gout, neuralgia,

rheumatism, insanity, and sleeplessness are among the conditions for which it is said to be beneficial. Its activities are virtually exclusively focused on the upper nerve centers (Elhendawy et al. 2019).

The antioxidant capacity and anticonvulsant activity of two nanoliposome-based *C. sativa* extracts and two nanostructured lipid-based *C. sativa* extracts were examined in electroshock convulsion models and i.v. PTZ and s.c. PTZ-induced seizure models. Comparing the four formulations to the *C. sativa* extract alone, they all demonstrated statistically significant increases in antioxidant activity. Furthermore, acquired data from in vivo investigations of the anticonvulsant activity in mice showed that all nanoformulations significantly increased latencies of myoclonic, clonic, and tonic seizures and increased the threshold of myoclonic twitch, generalized clonic seizures, and forelimb tonus (Mihailova et al. 2022).

β -Caryophyllene is a member of the cannabinoid family, which are ligands for the cannabinoid receptors (CB-Rs). CB-Rs, which are a form of metabotropic receptor, are essential for regulating neurotransmitters that control energy balance, metabolism, and immune functions. In contrast to conventional cannabinoids, β -caryophyllene has a distinct chemical structure and selectively activates the CB2-R (Francomano et al. 2019). It is well known that in mice models of neurological illness, CB2-R reduces neuroinflammation. More specifically, it has been demonstrated that administering CB2-R agonists lowers reactive gliosis and proinflammatory cytokine levels after neuronal damage. Because CB2-R activation reduces inflammation, CB2-R-targeting therapies may be especially effective for treating epilepsy diseases (Shapiro et al. 2019). A study evaluated the anticonvulsant effects of intranasal administration of a cationic nanoemulsion containing β -caryophyllene in PTZ-induced seizure models, given its poor oral bioavailability. Compared to the mice in the PTZ group, the intranasal nanoemulsion formulation exhibited an increased latency of tonic-clonic seizures, according to the in vivo examination. Furthermore, free β -caryophyllene given intranasally did not show any anticonvulsant effects (Nogueira et al. 2023).

Centella asiatica

The plant *Centella asiatica* Linn. is a member of the Apiaceae family. *Centella asiatica* is called "brain food" in India since it is one of the most effective herbs for healing wounds, curing skin conditions, and reviving nerve and brain cells. Since *Centella* offers so many health benefits, like being an antioxidant, an anti-inflammatory, a wound healer, and a memory enhancer, its use in food and drink has grown (Seevaratnam et al. 2012). Conceived as a revitalizing herb, *C. asiatica* has long been used, particularly in Ayurvedic and Chinese traditional medicine, to enhance memory, prevent

cognitive decline, and improve brain functions (Hanapi et al. 2021). *C. asiatica* contains four main bioactive compounds called centellosides: madecassoside, madecassic acid, asiaticoside, and asiatic acid (Kunjumon et al. 2022).

Researchers investigated the therapeutic potential of madecassoside-encapsulated alginate chitosan NPs in a rodent seizure model induced by PC. PC is an experimental model of epilepsy in the temporal lobe and causes a convulsive status epilepticus when injected intraperitoneally (i.p.). Formulated NPs reduced seizure severity and increased seizure latency in PC-induced seizure models. They observed that these NPs alleviated mRNA overexpression of oxidative and inflammatory mediators including mitogen-activated protein kinase (MAPK1), MAPK14, p53, Bcl-2, caspase 3, and heat shock protein family A (Hsp70) member 1B (hspa1b) (Kunjumon et al. 2021). In a different investigation, the anti-excitotoxicity and neuroprotective properties of asiaticoside-encapsulated alginate chitosan NPs were assessed in a mouse model of PC-induced seizures. By reducing the mRNA expression of inflammatory mediators (Bcl-2, caspase 3, hspa 1b, MAPK1, MAPK14, and p53) in the mouse brain, this formulation demonstrated reduced seizure severity and increased seizure latency and also relieved activity in the brains of mice challenged with PC (Kunjumon et al. 2022).

Crocus sativus

Crocus sativus is a member of the Iridaceae family and the Liliaceae line. Saffron has been used as a medication to enhance health and fight disease since antiquity, and it is also regarded as a food addition for taste, flavoring, and coloring, as well as its therapeutic effects. Saffron is used in traditional medicine as a diaphoretic, eupeptic, tranquilizer, expectorant, aphrodisiac, abortifacient, and emmenagogue, as well as to treat hepatic disorders, flatulence, spasm, vomiting, dental and gingival pain, insomnia, depression, seizures, cognitive disorders, lumbago, asthma, cough, bronchitis, colds, fever, cardiovascular disorders, and cancer (Moshiri et al. 2015; Samarghandian and Borji 2014).

Safranal appears to be a potential neuroprotective agent for treating epilepsy. In maximal electroshock-induced acute seizure models, safranal reduced convulsion percentage at the initial electroconvulsive threshold and significantly increased the electroconvulsive threshold. Research on PTZ-induced chronic seizure mice models demonstrated that safranal delayed the start of generalized seizures and lessened their intensity. Combining safranal with a nanostructured lipid vehicle (NLV) made its effects even more pronounced, comparable to the established anti-seizure medication sodium valproate. These findings suggest that safranal, especially when delivered via NLV, may offer a

valuable approach to managing generalized epilepsy (Bo-Qiang et al. 2018).

Cryptolepis sanguinolenta

Cryptolepis sanguinolenta is a species of flowering plant in the family Apocynaceae. A common antimalarial in West African ethnomedicine is *C. sanguinolenta* aqueous root extract (Ansah and Gooderham 2002). Among the main chemical components of *C. sanguinolenta* are flavonoids and tannins (Odoh and Akwuaka 2012), as well as alkaloids (six of which have been isolated from the root bark and identified spectroscopically as quindoline, cryptoquindoline, cryptolepine, biscryptolepine, 11-hydroxycryptolepine, and neocryptolepine) (Grellier et al. 1996; Kirby et al. 1995). *C. sanguinolenta* is traditionally used to treat various ailments including malaria (Grellier et al. 1996; Kirby et al. 1995) and bacterial infections (Gibbons et al. 2003), and also has anti-inflammatory and antioxidant effects (Jalali and Rahbardar 2023).

It has been reported that an NP formulation of cryptolepine called solid-lipid nanoparticle-cryptolepine (SLN-CRYP) significantly reduced the viability of *Taenia crassiceps* cysts and shrank cysticerci, resolving ventricular expansion and deviation. The treatment also inhibited hyperemia and was more effective than albendazole against microgliosis, calcification, edema, and meningitis. In addition, SLN-CRYP significantly reduced seizure severity and the frequency and duration of seizures in substance P-induced seizure models. Brain homogenate levels of interleukin 10 (IL-10) and interferon- α (IFN- α) were also significantly reduced with SLN-CRYP (Mante et al. 2021).

Curcuma longa

Curcuma longa is a member of the family, Zingiberaceae (Eigner and Scholz 1999; Omosa et al. 2017). The chemical compounds in turmeric are curcumin, essential oils, demethoxycurcumin, and bisdemethoxycurcumin which are responsible for many of its therapeutic effects (Li et al. 2011). *C. longa* is associated with a range of health benefits, including anti-inflammatory (Kocaadam and Şanlier 2017; Memarzia et al. 2021; Noorafshan and Ashkani-Esfahani 2013), antioxidant (Kocaadam and Şanlier 2017; Memarzia et al. 2021; Noorafshan and Ashkani-Esfahani 2013), anticancer (Kocaadam and Şanlier 2017; Memarzia et al. 2021), immunomodulatory properties (Memarzia et al. 2021), and positive effects on digestion and cognitive function (Hosseini and Hosseinzadeh 2018; Noorafshan and Ashkani-Esfahani 2013).

A study aimed to investigate the role of nitric oxide (NO) in the anticonvulsant effect of curcumin C3 complex NPs on PTZ-induced seizures in mice models. The anticonvulsant

effect of nanocurcumin was discovered to be dose dependent. The presence of L-arginine (an exogenous NO precursor) abolished nanocurcumin anticonvulsant action. In contrast, the addition of L-NG-Nitro arginine methyl ester (L-NAME), which is an inhibitor of inducible nitric oxide synthase (NOS) inhibitor, dramatically increased the sub-effective dose of nanocurcumin. Similar to aminoguanidine, a NOS inhibitor (Ygberg et al. 2021), nanocurcumin significantly raised seizure thresholds (Aminirad et al. 2017). In another study, in PTZ-induced kindling models, the neuroprotective effects of curcumin-loaded chitosan-alginate-sodium tripolyphosphate NPs have been demonstrated through the upregulation of erythropoietin (EPO) and klotho levels, as well as the downregulation of neuronal cell death and TNF- α messenger RNA (mRNA) expression (Mansoor et al. 2018). A study aimed to investigate the neuroprotective effects of curcumin-loaded nano-micelles in a rat model of status epilepticus induced by PC. The results demonstrated that curcumin NP exhibited potential in preventing cell death associated with proteins in the signaling cell pathway, oxidative stress, and neuroinflammation (Khadrawy et al. 2018). According to another report, the anticonvulsant properties of curcumin in the form of bovine serum albumin (BSA)-based nanocurcumin were used in a PTZ-induced seizure model. The intravenous PTZ paradigm was used to determine the threshold dose of PTZ for inducing clonic seizures, and the intraperitoneal PTZ paradigm was used to evaluate the latency to the onset of generalized clonus. The results showed a significant increase in the threshold and latency of clonic seizures. This effect was found to be better than that of natural curcumin (Moezi et al. 2021). In another study on Cur-SLNs, the in vitro experiments showed that Cur-SLNs were significantly more effective than free Cur in reducing neuronal apoptosis. Cur-SLNs decreased free radical levels and reversed mitochondrial dysfunction through the activation of the Bcl-2 family. The in vivo experiments demonstrated that SLNs were able to transport Cur across the BBB. In kainic acid-induced epileptic models treated with Cur-SLNs, an improvement in behavioral performance, with more healthy neurons and fewer apoptotic cells in the hippocampus, was observed. The in vivo mechanism was further explored, revealing that Cur-SLNs reduced neuronal apoptosis through the Bcl2 family and P38 MAPK pathways (Huang et al. 2020).

The anticonvulsant and neuroprotective effects of curcumin NPs are highlighted in the reviewed studies. Some of them show that curcumin NPs can raise seizure thresholds and latency, suggesting a strong anticonvulsant impact that seems to be dose dependent. In the meantime, the other studies highlight the neuroprotective action of curcumin NPs, demonstrating their efficacy in enhancing mitochondrial function and cellular signaling while lowering neuronal apoptosis, oxidative stress, and neuroinflammation. These

results imply that curcumin nanoparticles (NPs) have potential as a neuroprotective and anticonvulsant medication, providing dual benefits in seizure disorders.

Mimusops elengi

Mimusops elengi is an evergreen ornamental tree from the *Sapotaceae* family that has fragrant blossoms (Kadam et al. 2012). Alkaloids (Baliga et al. 2011), flavonoids (Shailajan and Gurjar 2015), triterpenoids (Baliga et al. 2011; Shailajan and Gurjar 2015), and phenolic compounds (gallic acid) (Shailajan and Gurjar 2015) are some of the chemical constituents that support its medicinal properties. Pre-clinical research has shown that its extract has antibacterial, anti-fungal, anti-decay, free radical inhibition, anti-blood sugar, anti-cancer, digestive protection, pain-relieving, and diuretic properties (Baliga et al. 2011; Karmakar et al. 2011).

A study used electroconvulsion and PTZ-induced clonic-type convulsion models in mice to examine the anticonvulsive properties of *M. elengi* silver nanoparticles (MESNPs). The MESNPs notably reduced the hind limb extension phase, shortened the duration of clonic-type seizures, and delayed the onset of clonic-type convulsions. Additionally, these NPs significantly decreased transfer latency. The analysis results indicated a substantial increase in the levels of GABA and acetylcholinesterase (AChE). Studies on antioxidants showed that MESNPs scavenged free radicals and markedly elevated antioxidant enzymes. Due to a considerable increase in antioxidant status, improved inhibitory neurotransmission, and suppression of excitatory neurohumoral transmission, MESNPs demonstrated strong anticonvulsive effects in both animals. MESNPs can change neurochemicals like GABA and AChE and reduce oxidative stress (Rajangam et al. 2022).

Nigella sativa

In many nations and civilizations, *Nigella sativa*, also referred to as black seeds, has long been used as a flavor and a healing substance for a variety of illnesses. Black seeds can be used to make herbal tea or as an active ingredient in herbal medications. In numerous studies, it has been demonstrated that *N. sativa* seeds, extracts, and oils can effectively manage a variety of conditions in both humans and model organisms, including diabetes, hypertension, oxidative stress, epilepsies, ulcers, asthma, inflammatory disorders, fatty liver, cancer, and arthritis (Majeed et al. 2021; Tavakoli et al. 2017). Thymoquinone is the primary bioactive substance found in black seeds (Ahmad et al. 2020a, b).

The anticonvulsant effect of the intranasal formulation of thymoquinone-loaded PLGA NPs was assessed in a study. It was shown that intranasal formulations with controlled release patterns and modest doses that crossed the BBB

successfully, increased the seizure threshold, and alleviated epilepsy with fewer side effects in an electroshock convulsion model (Ahmad et al. 2020a, b).

Passiflora caerulea

Passiflora caerulea, commonly known as blue passionflower, is a climbing plant known for its striking flowers and medicinal properties (Ulmer and MacDougal 2004). It has been used traditionally for its calming and soothing effects, particularly in herbal medicine (Harrison 2012). Most chemical substances in *Passiflora* species are polyphenolic compounds, including flavonoids (quercetin, rutin, apigenin, luteolin, chrysin, and C-glycosyl flavones, i.e., vitexin, isovitexin, orientin, and isoorientin), alkaloids, and caffeic acid (Frye and Hausteine 2007; Masteikova et al. 2008; Minteguiaga et al. 2021; Ożarowski and Karpiński 2021). *P. caerulea* has therapeutic effects, including anxiolytic, sedative, antioxidant, anti-inflammatory, and muscle relaxant properties (Deng et al. 2010; Minteguiaga et al. 2021; Ożarowski and Karpiński 2021).

In an in vivo study, the neuroprotective function of PLGA NPs loaded with chrysin against the PTZ-induced kindling chronic epilepsy model was investigated. The severity of the seizures was measured using the Racine scale, and the chrysin-loaded PLGA NPs showed the lowest score. The result showed that chrysin NPs reduced oxidative stress and neuronal apoptosis and increased the expression of antioxidant enzymes and transcription factors in a model of epilepsy (Zhang et al. 2021b).

Piper nigrum

Traditional medicine has employed black pepper, a member of the Piperaceae family, to cure various illnesses. The existence of metabolites such as phenolic chemicals, alkaloids, flavonoids, carotenoids, and terpenoids gives black pepper its pharmacological potential. Black pepper's primary bioactive ingredient, piperine, has several medicinal uses (Ashokkumar et al. 2021).

In an in vivo study, it has been demonstrated that chitosan-sodium tripolyphosphate NPs loaded with piperine increased the latency of generalized clonic-tonic seizures, reduced their duration in fully kindled animals, and decreased seizure severity in a PTZ-induced kindling model of epilepsy. They also enhanced neuroprotection and decreased cell loss and astrocyte activation (Anissian et al. 2018). Another study evaluated the therapeutic efficacy of piperine-loaded Spanlastics (PIP-SPL) for the treatment of epilepsy through in vivo evaluations in the PTZ-induced seizure model. The results indicated that the PIP-SPL formulation exhibited a superior anticonvulsant effect compared to standard diazepam treatment. Improvements in seizure

activity (kindling percentage and seizure score decrease and seizure latency increase), neuromuscular coordination, and oxidative biomarker activity evidenced this. The study suggests that PIP-SPL formulation holds promise as an intranasal delivery method for managing epilepsy (Gupta et al. 2023). In another investigation, piperine drug molecules encapsulated in copper oxide quantum dots (CuQDs) coated hyaluronic acid (HA)/PLGA nanovesicles were shown to provide a unique platform for sustained and efficient drug delivery for an antiepileptic effect through in vitro and in vivo tests. According to the cytotoxicity data, CuQDs@HA/PLGA has a low level of toxicity in brain microvascular endothelial cells. The synthesized NPs inhibit seizure activity and increase the latency of generalized clonic-tonic seizures and myoclonic jerks in an in vivo PTZ-induced seizure model (Zhu et al. 2020).

Piperine encapsulation in nanoformulations can enhance blood–brain transport and target delivery, providing an efficient therapeutic approach and creating novel opportunities for epilepsy treatment.

***Physalis peruviana* (Cape gooseberries)**

Physalis peruviana is a species of plant in the nightshade family *Solanaceae*. *P. peruviana* is an annual in temperate locations, but a perennial in the tropics (Fischer et al. 2011). The ethanolic extract of *P. peruviana* fruits contains valuable and active compounds such as flavonoids, tannins, carotenoids, phenols, alkaloids, and vitamins C, B₃, and B₆ (El-Beltagi et al. 2019). Therapeutic effects are associated with several health benefits, such as antioxidant, anti-inflammatory, immune support, digestive health, and potential anticancer effects (Wu et al. 2005).

In an in vivo study in the PTZ-induced seizure model, pretreatment with Rutin-loaded selenium NPs decreased seizure severity and duration and increased the latency time. The efficacy of these NPs was demonstrated in reducing Bax and caspase-3 levels, as well as levels of malondialdehyde (MDA) and NO. Additionally, it was found that these NPs increased the levels of Bcl-2, as well as antioxidant enzymes such as superoxide dismutase (SOD), catalase, glutathione (GSH), Nrf2, and HO-1. Furthermore, the nanotherapy effectively reduced the levels of inflammatory mediators (TNF- α , IL-6, COX-2, and nuclear factor- κ B (NF- κ B)) in the hippocampal tissues of mice (Mohamed et al. 2023).

Rosa webbiana

Rosa webbiana, commonly known as the Himalayan rose, is a species of rose native to the Himalayan region. It is valued for its ornamental beauty as well as its medicinal properties (Hummer and Janick 2009). Rose hips contain numerous bioactive substances, including amino acids, anthocyanins,

ascorbic acid, flavonoids, pectins, phenolic acids, tannins, and tocopherols (Shameh et al. 2018; Sharma et al. 2023). Various parts of *Rosa* species have shown multiple bioactive properties, including antioxidant, anticancer, anti-obesity, antiestrogenic, anticonvulsant, and antidiabetic effects (Firdous et al. 2021; Shameh et al. 2018; Sharma et al. 2023).

A study showed the potential neuroprotective effects of NPs of *n*-hexane extract derived from *R. webbiana*. Although diazepam exhibited a stronger anticonvulsive effect, the NPs also demonstrated anticonvulsive properties by reducing seizure severity in the PTZ-induced epilepsy model. The results suggest that the extract may have a downregulatory effect on the elevated levels of phosphorylated NF- κ B and TNF- α observed in epilepsy. These findings indicate that the enhanced anticonvulsant activity of *R. webbiana* could be responsible for its neuroprotective effects. These results suggest that they provide valuable insights into the potential use of *R. webbiana* extract as a natural and effective treatment option for epilepsy and related conditions (Firdous et al. 2021).

Table 3 shows the characteristic items (size, polydispersity index (PDI), zeta potential, loading efficacy, and release) of different plant NPs and Table 4 shows the effects of herbal NPs on epilepsy.

Other herbal compounds

Ellagic acid

Ellagic acid (EA) is a type of polyphenol present in many fruits and vegetables. It is the dilactone form of hexahydroxydiphenic acid (Bodoira and Maestri 2020). This compound is also found in edible nuts and oak species like the North American white oak (*Quercus alba*) and the European red oak (*Quercus robur*) (Mämmelä et al. 2000). Therapeutic effects of ellagic acid include antioxidant activity, anti-inflammatory properties, anticancer effects, cardiovascular health, antimicrobial properties, liver protection, and skin health (Ríos et al. 2018; Yılmaz et al. 2016; Zhang et al. 2014).

In a study, male mice that were given PTZ to produce seizures were used to assess the neuroprotective and anticonvulsant effects of EA encapsulated in calcium-alginate (Ca^{2+} -ALG) NPs. Throughout the trial, Ca^{2+} -ALG NPs avoided seizures by decreasing the percentile occurrence of convulsions and had a greater impact than free EA. Ca^{2+} -ALG NPs mitigated elevated serum and brain levels of homocysteine and amyloid- β while also preventing the increase in glutamate and lowering GABA concentrations. Ca^{2+} -ALG NPs improved elevated IL-6 and TNF- α better than free EA. PTZ-induced oxidative stress was lessened by Ca^{2+} -ALG NPs, as seen by higher brain GSH, glutathione

reductase (GR), and glutathione peroxidase (GPx) levels and lower 4-hydroxynonenal (4HNE) levels. Apoptosis and its regulatory proteins, such as cytochrome *c*, p53, Bax, Bcl2, and caspase-3 and caspase-9, improved in tandem with these modifications (El-Missiry et al. 2020).

Quercetin

Several plants contain quercetin, a flavonoid molecule with a wide range of biological activity. Studies on quercetin have indicated that it may have therapeutic uses. Studies on humans and animals have documented a variety of quercetin's pharmacological effects, including blood pressure reduction, cardiovascular protection, weight loss, improved hyperglycemia, hypolipidemia, antioxidant, anti-inflammatory, anti-cancer, and neuroprotective properties (Grewal et al. 2021; Hosseini et al. 2021; Yang et al. 2020).

A study investigated the synthesis and evaluation of novel quercetin-conjugated Fe_3O_4 - β -cyclodextrin NPs. The results show that quercetin was able to bind to these NPs firmly. Although quercetin-conjugated NPs did not show a significant reduction in seizure stage compared to saline or blank NP, they demonstrated noticeably higher efficacy than free quercetin in terms of seizure activity reduction (increase in latency of generalized clonic-tonic seizures and myoclonic jerks), neuronal damage prevention, and astrocyte activation reduction in a PTZ-induced kindling model (Hashemian et al. 2019).

Tannic acid

Tannic acid (TNA) is the most active component among polyphenols obtained from plants. TNA is primarily found in the twigs of certain trees, Tara pods (*Caesalpinia spinosa*), green tea, gallnuts (*Quercus infectoria* or *Rhus semialata*), grapes, leaves of Sicilian sumac (*Rhus coriaria*), and oak. Previous reports indicate that TNA exhibits antiepileptic effects, anti-cancer properties, antiviral and antimicrobial activities, as well as antioxidant effects. It also provides significant protection against various diseases and their associated damage (Andrade Jr et al. 2006; Wang et al. 2016).

In a study, the brain bioavailability of chitosan-coated TNA-vit E-loaded PLGA NPs was assessed through intranasal delivery. The PLGA NPs were created using an emulsion solvent evaporation method. Coating the NP with chitosan improved antiepileptic efficacy, brain targeting, and drug release. The results showed a significant improvement in brain bioavailability of TNA in treated epilepsy models, with no observed mortalities or morphological changes in the brain or nasal mucosa tissues. The NPs also reduce generalized clonic-tonic seizures and myoclonic jerks in electroshock and PTZ-induced epilepsy models (Ahmad et al. 2022).

Table 3 Characterization of different plant NPs

Sample	Characterization		Zeta potential (mV)	Loading efficacy	Release	References
	Size (nm)	PDI				
<i>Bacopa monnieri</i>						
PLGA-b-PEG nano-encapsulated bacopside A	165.5	0.044	– 32.5	N/A	N/A	(Sekhar et al. 2024)
PLGA-b-PEG nano-encapsulated bacopside I	689.6	0.514	– 5.59	N/A	N/A	(Sekhar et al. 2024)
<i>Berberis vulgaris</i>						
Berberine-loaded zein/hyaluronic acid composite	297.2 ± 1.86	0.144 ± 0.009	– 39 ± 0.003	83.75 ± 1.39%	84.23 ± 2.34% through 48 h	(El-Nahas et al. 2023)
<i>Camellia sinensis</i>						
Chitosan-coated catechin-loaded PLGA NPs 1 (2% chitosan)	106.31 ± 2.91	0.239 ± 0.008	21.64 ± 1.04	80.36 ± 4.01%	81.34 ± 4.02% through 24 h	(Ahmad et al. 2020a, b)
Chitosan-coated catechin-loaded PLGA NPs 2 (4% chitosan)	142.43 ± 3.94	0.364 ± 0.023	24.34 ± 1.59	81.66 ± 4.61%		
<i>Cannabis sativa</i>						
<i>Cannabis sativa</i> extract–loaded nanoliposome 1	130.70 ± 2.526	0.285 ± 0.024	– 28.90 ± 3.269	99.86 ± 1.01%	≈52% through 24 h in PBS medium	(Mihalova et al. 2022)
<i>Cannabis sativa</i> extract–loaded nanoliposome 2	159.80 ± 3.230	0.253 ± 0.017	– 43.8 ± 2.387	99.89 ± 0.76%	≈39% through 24 h in PBS medium	(Mihalova et al. 2022)
<i>Cannabis sativa</i> extract–loaded nanostructured lipid 1	111.80 ± 1.256	0.213 ± 0.053	– 24.60 ± 0.763	98.02 ± 0.62%	≈60% through 24 h in PBS medium	(Mihalova et al. 2022)
<i>Cannabis sativa</i> extract–loaded nanostructured lipid 2	133.40 ± 2.319	0.202 ± 0.047	– 31.20 ± 1.117	87.19 ± 0.42%	≈60% through 24 h in PBS medium	(Mihalova et al. 2022)
Cationic nanoemulsion containing β-caryophyllene	244.03 ± 15.67	0.18 ± 0.04	41.1 ± 1.57	98%	N/A	(Nogueira et al. 2023)
<i>Centella asiatica</i>						
Madecassoside-loaded alginate chitosan NPs	200–600	0.431	– 32.2	≥ 90%	10% through 24 h in PBS medium	(Kunjumon et al. 2021)
Asiaticoside-loaded alginate chitosan NPs	200–400	0.567	– 14.1	≥ 90%	10% through 24 h in PBS medium	(Kunjumon et al. 2022)
<i>Crocus sativus</i>						
NLV carried safranal	164 ± 43.2	0.137	– 23.2 ± 1.8	N/A	N/A	(Bo-Qiang et al. 2018)
<i>Cryptolepis sanguinolenta</i>						
Cryptolepine solid-lipid NPs	226.4 ± 8.0	0.19	– 17.2 ± 1.3	81.6 ± 2.7%	79% through 12 h in PBS medium	(Mante et al. 2021)
<i>Curcuma longa</i>						
Curcumin C3 complex NPs	N/A	N/A	N/A	N/A	N/A	(Aminirad et al. 2017)
Curcumin-loaded chitosan-alginate-sodium tripolyphosphate NPs	N/A	N/A	N/A	N/A	N/A	(Mansoor et al. 2018)
Curcumin-loaded nano-micelle	33 to 44	N/A	N/A	N/A	N/A	(Khadravy et al. 2018)
BSA-based nanocurcumin	N/A	N/A	N/A	N/A	N/A	(Moezi et al. 2021)
Cur-SLNs	50	N/A	– 27.16 ± 1.65	N/A	N/A	(Huang et al. 2020)
<i>Mimusops elengi</i>						
<i>M. elengi</i> silver NPs	24 nm	N/A	–29.5 mV	N/A	N/A	(Rajangam et al. 2022)

Table 3 (continued)

Sample	Characterization		Release		References
	Size (nm)	PDI	Zeta potential (mV)	Loading efficacy	
<i>Nigella sariva</i>					
Thymoquinone-loaded PLGA NPs	97.36 ± 2.01	0.263 ± 0.004	− 17.98 ± 1.09	82.49 ± 2.38%	77.58 ± 3.84% through 24 h (Ahmad et al. 2020a, b)
<i>Passiflora caerulea</i>					
Chrysin-loaded poly (lactic-co-glycolic acid) NPs	115.79 ± 2.84	0.152 ± 0.006	− 29.20 ± 0.24	93.27 ± 1.2%	72.05 ± 3.71% through 22 h (Zhang et al. 2021a, b)
<i>Piper nigrum</i>					
Piperine-loaded chitosan-sodium tripolyphosphate	30	N/A	N/A	53%	N/A (Anissian et al. 2018)
Piperine-loaded Spanlastics	152.4 ± 2.98	0.1118 ± 0.009	− 36.83	72.93 ± 2.09%	82.32% through 24 h (Gupta et al. 2023)
Piperine-loaded copper oxide quantum dots coated with hyaluronic acid and PLGA	10	N/A	N/A	85.43 ± 5.0%	45% in PBS medium and 35% in acidic medium through 7 days (Zhu et al. 2020)
<i>Physalis peruviana</i>					
Rutin-loaded selenium NPs	≈ 70	N/A	− 23.3	N/A	N/A (Mohamed et al. 2023)
<i>Rosa webbiana</i>					
<i>Rosa webbiana</i> (n-hexane cs-NPs)	N/A	0.069	4.62	69%	N/A (Firdous et al. 2021)

Herbal NPs are a promising improvement in epilepsy treatment, with various advantages over regular ASMs. Herbal NPs, compared to conventional drugs, are capable of improving the bioavailability and targeted delivery of medicinal chemicals derived from plants, resulting in a variety of adverse effects and limited efficacy. By focusing on reducing systemic exposure and potential negative effects, the treatment becomes safer and more effective. Furthermore, the inherent anti-inflammatory and neuroprotective qualities of herbal extracts can work in tandem with NP formulations, potentially improving seizure control and cognitive results.

Table 5 shows the characteristic items (size, PDI, zeta potential, loading efficacy, and release) of different herbal compound NPs and Table 6 shows the effects of herbal compound NPs on epilepsy.

Conclusion

The review manuscript concludes with evidence supporting the potential of herbal NPs as treatments for epilepsy. These plant NPs not only mitigate oxidative stress and neuronal apoptosis but also enhance the expression of protective proteins such as HO-1 and NAD(P)H. The findings underscore the capacity of herbal NP therapies to modulate seizure activity and provide neuroprotection through various mechanisms. Specifically, they reduce levels of reactive nitrogen species (NO) and oxidative stress via the Nrf2/antioxidant response element/HO-1 signaling pathway. Additionally, these NPs decrease neurodegeneration and neuroinflammation by lowering the expression of pro-inflammatory markers such as COX-2, IL-10, and IFN- α , as well as TNF- α and NF- κ B. This results in a reduction of seizure behaviors, neuronal loss, glial activation, and overall cell death, leading to improved neuromuscular coordination and favorable changes in oxidative biomarkers.

Moreover, the nanoformulation of herbal medicines enhances their anticonvulsant effects by increasing levels of GSH, GR, and GPx in the brain while decreasing NO levels and the activity of enzymes like AChE and Na⁺/K⁺-ATPase in critical regions such as the cortex and hippocampus. This modulation also elevates GABA concentrations and addresses the imbalance between excitation and inhibition in neuronal activity.

While these promising results highlight the multifaceted benefits of herbal NPs such as enhanced bioavailability, targeted delivery, sustained release, reduced toxicity, and enhanced penetration of the BBB of herbal extracts, further research is essential to validate their efficacy and safety in clinical settings.

These NPs showed promising anticonvulsant activity, with the most effective compounds reducing seizure frequency and severity, indicating their potential for further

Table 4 The effects of different plant NPs on epilepsy

Compound	Study model	Doses and duration	Results	References
<i>Bacopa monnieri</i> PLGA-b-PEG nano-encapsulated bacoside A and bacopaside I	In vitro, mouse neuronal stem cells (mNSC) and neuroblastoma cells (N2a) In vivo, male Wistar rats	5, 25, 50, 100, 200, and 500 µg/mL 50 µg/mL, i.v	↑ Duration of NREM sleep ↓ Sleep latency and epileptic spikes ↓ Cytotoxicity when exposed to kainic acid ↓ Apoptosis	(Sekhar et al. 2024)
<i>Berberis vulgaris</i> Berberine-loaded zein/hyaluronic acid composite	In vivo, adult male Sprague–Dawley rats	4 mg/kg, 6 days, s.c	↓ Seizure severity and percentage of animals that developed status epilepticus ↑ Latency of status epilepticus ↑ Antioxidant activity ↓ Neuro-degeneration ↓ Expression of COX-2 and TNF-α	(El-Nahas et al. 2023)
<i>Camellia sinensis</i> Chitosan-coated catechin-loaded PLGA NPs	In vivo, albino rats	10 mg/kg, i.n	↑ Hind limb tonic extension threshold ↓ Post-hind limb tonic extension recovery time	(Ahmad et al. 2020a, b)
<i>Cannabis sativa</i> <i>C. sativa</i> extract-loaded nanoliposomes and nanostructured lipids	In vivo, male ICR mice	30, 100, and 300 mg/kg, i.p	↑ Latencies of myoclonic, clonic, and tonic seizures ↑ The threshold of myoclonic twitch, generalized clonic seizures, and forelimb tonus	(Mihailova et al. 2022)
Cationic nanoemulsion containing β-caryophyllene <i>Centella asiatica</i> Madecassoside-loaded alginate chitosan NPs	In vivo, male Wistar rats In vivo, Swiss albino mice	40 mg/mL/min bilateral i.n. infusion 20 mg/kg, i.v	↑ Latencies of tonic–clonic seizures ↓ Seizure severity ↑ Seizure latency ↓ mRNA expression of MAPK1, MAPK14 genes together with p53, Bcl-2, caspase 3, and hspa1b	(Nogueira et al. 2023) (Kunjumon et al. 2021)
Asiaticoside-loaded alginate chitosan NPs	In vivo, Swiss albino mice	20 mg/kg, i.v	↓ Seizure severity ↑ Seizure latency ↓ mRNA expression of MAPK1, MAPK14, p53, Bcl-2, caspase 3, and hspa1b	(Kunjumon et al. 2022)
<i>Crocus sativus</i>				

Table 4 (continued)

Compound	Study model	Doses and duration	Results	References
NLV-carried safranal	In vivo, mice	100 mg/kg, i.p	↓ Convulsion percentage ↑ Electroconvulsive threshold ↓ Onset of generalized seizures and their severity ↓ Seizure-related neural damage	(Bo-Qiang et al. 2018)
<i>Cryptolepis sanguinolenta</i> Cryptolepine solid-lipid NPs	In vivo, female BALB/c mice	5 mg/kg, 28 days, i.g	↓ Seizure severity ↓ Seizure frequency and duration ↓ Eliminated cysticerci and resolved ventricular expansion and hyperemia ↓ Microgliosis, calcification, and edema ↓ Levels of IL-10 and IFN- α	(Mante et al. 2021)
<i>Curcuma longa</i> Curcumin C3 complex NPs	In vivo, male NMRI albino mice	1, 5, 10, 20, 40, and 80 mg/kg, i.p	↑ Seizure thresholds, non-selective NOS inhibitor, and selective iNOS inhibitor ↓ NO ↑ Levels of EPO and klotho ↓ Neuronal cell death, mRNA level of TNF- α	(Aminirad et al. 2017) (Mansoor et al. 2018)
Curcumin-loaded chitosan-alginate-sodium tripolyphosphate NPs	In vivo, male NMR mice	12.5 mg/kg, 10 days, i.p	↓ Lipid peroxidation and levels of TNF- α and caspase-3 ↑ GSH and NO and activities of (AChE) and Na ⁺ /K ⁺ -ATPase in the cortex and hippocampus	(Khadrawy et al. 2018)
Curcumin-loaded nano-micelles	In vivo, male Wistar rats	50 mg/kg, 4 days, i.p	↑ Threshold and latency of clonic seizure ↑ Latency of generalized clonus	(Moezi et al. 2021)
BSA-based nanocurcumin	In vivo, male NMRI mice	50 and 100 mg/kg, p.o	↓ Free radical levels ↓ Mitochondrial dysfunction through the activation of the Bcl-2 family	(Huang et al. 2020)
Cur-SLNs	In vitro, cortical neurons In vivo, C57 mice	1 and 5 μ g/mL, for 6 h 20 mg/kg, i.p	↑ Behavioral performance ↓ Apoptosis through the Bcl2 family and p38 MAPK pathways	
<i>Minusops elengi</i> <i>M. elengi</i> silver NPs	In vivo, adult albino mice	200 mg/kg, p.o	↓ Hind limb extension phase, the duration of clonic-type seizures ↑ Latency of clonic-type convulsions ↑ Levels of GABA and AChE and antioxidant status, improved inhibitory neurotransmission	(Rajangam et al. 2022)
<i>Nigella sativa</i> Thymoquinone-loaded PLGA NPs	In vivo, albino rats	10 mg/kg, i.n	↑ Seizure threshold	(Ahmad et al. 2020a, b)

Table 4 (continued)

Compound	Study model	Doses and duration	Results	References
<i>Passiflora caerulea</i> Chrysin-loaded poly (lactic-co-glycolic acid) NP	In vivo, male adult Wistar rats	5 and 10 µg/mL, every second day to reach total doses of 22, p.o	↓ Seizure severity ↑ Nuclear Nrf2, HO-1, and NAD(P)H quinone oxidoreductase 1 ↓ Oxidative stress and neuronal apoptosis	(Zhang et al. 2021a, b)
<i>Piper nigrum</i> Piperine-loaded chitosan-sodium tripolyphosphate	In vivo, male NMR mice	5 and 10 mg/kg, 10 days, i.p	↑ The latency of generalized clonic-tonic seizures ↓ Seizure duration and severity ↓ Cell loss and astrocyte activation	(Anissian et al. 2018)
Piperine-loaded Spanlastics	In vivo, Wistar albino rats	1 mg/mL, i.n	↓ Kindling percentage and seizure score ↑ Seizure latency Improvements in neuromuscular coordination and oxidative biomarker activity	(Gupta et al. 2023)
Piperine-loaded copper oxide quantum dots coated with hyaluronic acid and PLGA	In vivo, male Wistar rats	Different concentrations	↓ Seizure activity ↑ Latency of generalized clonic-tonic seizures and myoclonic jerks	(Zhu et al. 2020)
<i>Physalis peruviana</i> Rutin loaded with selenium NPs	In vivo, male Swiss mice	100 mg/kg, 10 days, p.o	↑ Latency time and AchE, brain-derived neurotrophic factor, dopamine, and norepinephrine levels ↑ SOD, catalase, GSH, Nrf2, and HO-1, and also levels of Bcl-2 ↓ Severity and duration of seizures ↓ MDA and NO levels ↓ Bax and caspase-3 ↓ TNF-α, IL-6, COX-2, and NF-κB	(Mohamed et al. 2023)
<i>Rosa webbiana</i> n-Hexane extract of <i>R. webbiana</i> fruit NPs	In vivo, male Sprague–Dawley rats	50, 150, and 300 mg/kg Single dose	↓ Seizure severity ↓ Neuro-inflammation, p-TNF-α and p-NF-κB	(Firdous et al. 2021)

Table 5 Characterization of herbal compound NPs

Sample	Characterization				Release	References
	Size (nm)	PDI	Zeta potential (mV)	Loading efficacy		
Ellagic acid-loaded Ca^{2+} -ALG NPs	670 ± 4.2	N/A	-27 ± 2.6	N/A	$\approx 100\%$ through 8 h	(El-Missiry et al. 2020)
Quercetin-conjugated Fe_3O_4 - β -cyclodextrin NPs	< 50	N/A	N/A	N/A	N/A	(Hashemian et al. 2019)
Chitosan-coated tannic acid-vit E-loaded PLGA NPs	105.7 ± 11.02	0.169 ± 0.002	-22.3 ± 2.3	$69.31 \pm 5.89\%$	$78.38 \pm 5.08\%$ through 24 h	(Ahmad et al. 2022)

Table 6 The effects of herbal compound NPs on epilepsy

Compound	Study model	Doses and duration	Results	References
Ellagic acid-loaded Ca^{2+} -ALG NPs	In vivo, male Swiss albino mice	50 mg/kg, every other day for 33 days, p.o	↓ Percentile occurrence of convulsions ↓ Glutamate and amyloid- β and homocysteine levels ↑ GSH, GR, and GPx levels ↑ GABA concentrations ↓ 4HNE levels induced amelioration of apoptosis and its regulating proteins, including cytochrome c, p53, Bax, Bcl2, and caspase-3 and caspase-9 ↓ IL-6 and TNF- α	(El-Missiry et al. 2020)
Quercetin-conjugated Fe_3O_4 - β -cyclodextrin NPs	In vivo, male NMRI mice	25 and 50 mg/kg, 10 days, i.p	↑ Latency of generalized clonic-tonic seizures and myoclonic jerks ↓ Neuronal loss and astrocyte activation	(Hashemian et al. 2019)
Chitosan-coated tannic acid-vit E-loaded PLGA NPs	In vivo, Wistar rats	15 mg/kg, i.n. and i.v	↓ Generalized clonic-tonic seizures and myoclonic jerks No toxicological results	(Ahmad et al. 2022)

development in epilepsy treatment. However, only those with strong anticonvulsant effects should be considered for treating epilepsy. Many of the NPs mainly displayed neuroprotective and biochemical properties, which, while helpful, may serve as complementary treatments by protecting neurons, reducing inflammation, and alleviating oxidative stress associated with seizures. Further research is essential to validate their efficacy and safety in clinical settings. This exploration will be crucial in establishing herbal NPs as a complementary treatment or superior alternative to conventional anti-epileptic treatments, potentially leading to more effective and safer management strategies for epilepsy (Fig. 4).

Future directions

1. Enhancing bioavailability: Future research should focus on improving the bioavailability of herbal anti-convulsants through advanced NP formulations. This could involve optimizing the size, surface properties, and composition of NPs to ensure better absorption and therapeutic concentrations in the brain.
2. Targeted drug delivery systems: The development of NPs specifically engineered for targeted delivery to affected brain regions can enhance the efficacy of herbal treatments while minimizing systemic side effects. This approach could significantly improve treatment outcomes for epilepsy patients.

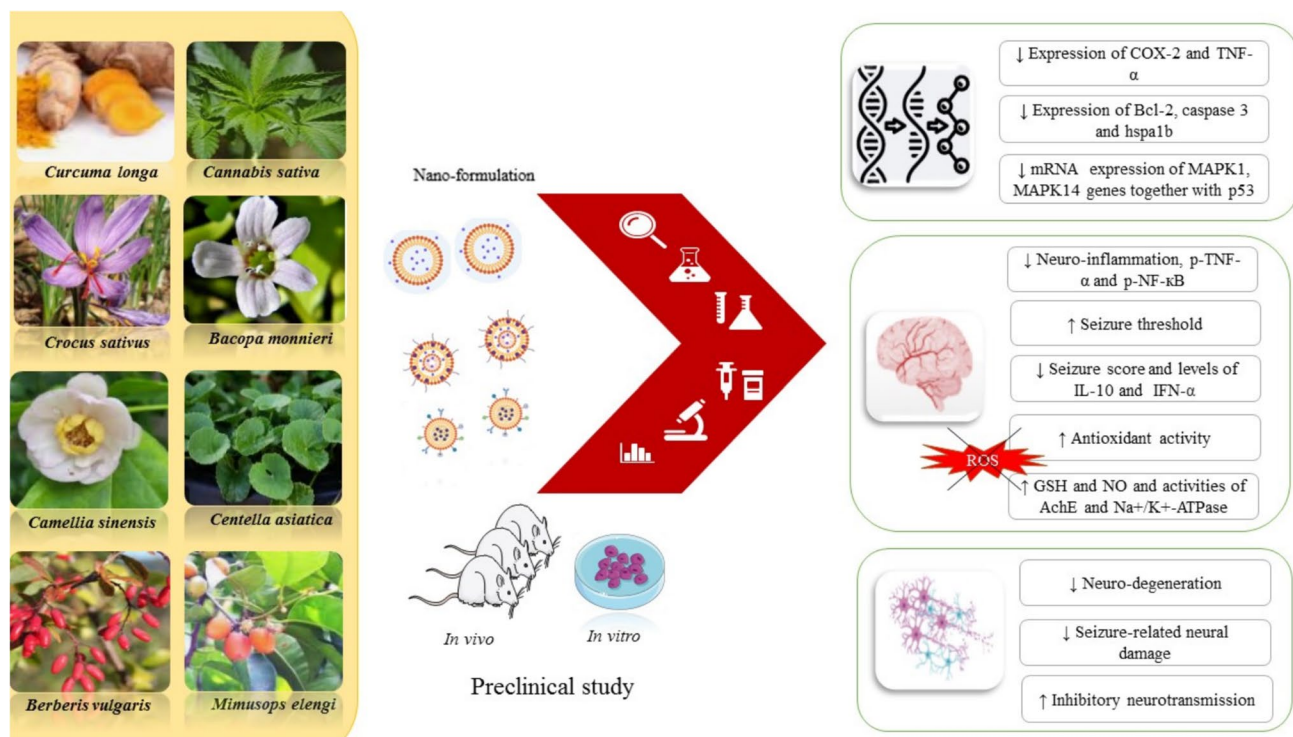


Fig. 4 Overview of the mechanisms by which nanoparticles exert their anti-epileptic effects

- Long-term efficacy studies: Conducting long-term studies on the efficacy and safety of herbal NPs in diverse populations is essential. This would help establish their potential as a sustainable treatment option for epilepsy, particularly for patients who are resistant to conventional therapies.
- Mechanistic studies: Further investigation into the mechanisms of action of herbal NPs is needed. Understanding how these compounds modulate neurotransmitter systems, ion channels, and oxidative stress could lead to more effective combination therapies for epilepsy management.
- Overcoming drug resistance: Research should explore how NP formulations can overcome drug resistance mechanisms that limit the effectiveness of traditional antiepileptic drugs. This could involve studying interactions between NPs and cellular pathways associated with drug resistance.
- Regulatory frameworks and standardization: Establishing regulatory guidelines and standardization protocols for the production and testing of herbal NPs is crucial. This will ensure safety, efficacy, and consistency in their application as therapeutic agents for epilepsy.
- Integration with conventional treatments: Future directions should include exploring the synergistic effects of herbal NPs when used in conjunction with conventional antiepileptic drugs. This could enhance overall treatment

efficacy and provide a holistic approach to managing epilepsy symptoms.

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Author contribution HFK: Writing—original draft. AMZ: Writing—review & editing. HH: Conceptualization, Methodology, Supervision. All authors read and approved the final manuscript. The authors declare that all data were generated in-house and that no paper mill was used. During the preparation of this work, the authors used AI-assisted technologies to rephrase to reduce plagiarism and improve language and grammar. After using this tool/service, the authors reviewed and edited the content as needed and take(s) full responsibility for the content of the publication. The authors confirm that no paper mill and artificial intelligence was used.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article, and also H.H. is the Editor of *Naunyn-Schmiedeberg's Archives of Pharmacology* and followed the submission guidelines of the journal (https://link.springer.com/journal/210/submission-guidelines#Instructions%20for%20Authors_Competing%20Interests).

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